Van Lang University · Faculty of Information Technology · Mobile Application Programming

LAB 8 · DEBUGGING REPORT (BUGGYREADER)

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GitHub: https://github.com/NeitLN/Lab8_NetworkJSON_2474802010391 (folder BuggyReader)

1. Steps

BuggyReader was created with the code from the handout and run on a Pixel 7 emulator (Android 14, API 34). Tapping "Load posts" crashes the app. The trace was taken from Logcat (level Error), from FATAL EXCEPTION down to the first `com.vlu.buggyreader` line. After fix 1 the app was run again and a different crash appeared, then fix 2 was applied. After fix 2 the app shows all 100 titles and does not crash anymore.

Commits: `ef641c7` original buggy code, `b48975a` fix 1, `56b9b2b` fix 2.



After 2 fixes

2. Debugging table

#	Exception class	Key Logcat line	Root cause (own words)	Fix applied (file and line)	AI used? (prompt and what you changed)
1	`android.os.NetworkOnMainThreadException`	`at com.vlu.buggyreader.MainActivity.loadPosts(MainActivity.kt:25)`	loadPosts() runs right inside the button click, so the HTTP request is made on the main (UI) thread. Android does not allow network calls on this thread because the screen would freeze while waiting and the app could get an ANR.	MainActivity.kt line 28 to 40: use `lifecycleScope.launch` and `withContext(Dispatchers.IO)` (line 32), the request is moved to `downloadTitles()` (line 43 to 59), the TextView is updated on main (line 33). build.gradle.kts line 42 to 43: add lifecycle and coroutines.	ChatGPT explained the error. I moved the request off the UI thread and reran the app.
2	`java.lang.SecurityException`	`Permission denied (missing INTERNET permission?)` at `MainActivity.downloadTitles(MainActivity.kt:48)`	The manifest does not declare the INTERNET permission, so the app cannot open a connection to the server. This error only shows up after fix 1, because before that the app already crashed at the main thread check.	AndroidManifest.xml line 6: add `<uses-permission android:name="android.permission.INTERNET" />` inside `<manifest>`, outside `<application>`.	I added the permission and verified 100 titles loaded.

3. Stack traces from Logcat

Error 1 (first run, tap Load posts):

```
FATAL EXCEPTION: main
Process: com.vlu.buggyreader, PID: 25636
android.os.NetworkOnMainThreadException
    at android.os.StrictMode$AndroidBlockGuardPolicy.onNetwork(StrictMode.java:1675)
    at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:115)
    at java.net.Inet6AddressImpl.lookupAllHostAddr(Inet6AddressImpl.java:103)
    at java.net.InetAddress.getAllByName(InetAddress.java:1152)
    at com.android.okhttp.Dns$1.lookup(Dns.java:41)
    at com.android.okhttp.internal.http.RouteSelector.resetNextInetSocketAddress(RouteSelector.java:178)
    at com.android.okhttp.internal.http.RouteSelector.nextProxy(RouteSelector.java:144)
    at com.android.okhttp.internal.http.RouteSelector.next(RouteSelector.java:86)
    at com.android.okhttp.internal.http.StreamAllocation.findConnection(StreamAllocation.java:176)
    at com.android.okhttp.internal.http.StreamAllocation.findHealthyConnection(StreamAllocation.java:128)
    at com.android.okhttp.internal.http.StreamAllocation.newStream(StreamAllocation.java:97)
    at com.android.okhttp.internal.http.HttpEngine.connect(HttpEngine.java:289)
    at com.android.okhttp.internal.http.HttpEngine.sendRequest(HttpEngine.java:232)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.execute(HttpURLConnectionImpl.java:465)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponse(HttpURLConnectionImpl.java:411)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getInputStream(HttpURLConnectionImpl.java:248)
    at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getInputStream(DelegatingHttpsURLConnection.java:211)
    at com.android.okhttp.internal.huc.HttpsURLConnectionImpl.getInputStream(HttpsURLConnectionImpl.java:30)
    at com.vlu.buggyreader.MainActivity.loadPosts(MainActivity.kt:25)
```

Error 2 (after fix 1, tap Load posts again):

```
FATAL EXCEPTION: main
Process: com.vlu.buggyreader, PID: 26251
java.lang.SecurityException: Permission denied (missing INTERNET permission?)
    at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:150)
    at java.net.Inet6AddressImpl.lookupAllHostAddr(Inet6AddressImpl.java:103)
    at java.net.InetAddress.getAllByName(InetAddress.java:1152)
    at com.android.okhttp.Dns$1.lookup(Dns.java:41)
    at com.android.okhttp.internal.http.RouteSelector.resetNextInetSocketAddress(RouteSelector.java:178)
    at com.android.okhttp.internal.http.RouteSelector.nextProxy(RouteSelector.java:144)
    at com.android.okhttp.internal.http.RouteSelector.next(RouteSelector.java:86)
    at com.android.okhttp.internal.http.StreamAllocation.findConnection(StreamAllocation.java:176)
    at com.android.okhttp.internal.http.StreamAllocation.findHealthyConnection(StreamAllocation.java:128)
    at com.android.okhttp.internal.http.StreamAllocation.newStream(StreamAllocation.java:97)
    at com.android.okhttp.internal.http.HttpEngine.connect(HttpEngine.java:289)
    at com.android.okhttp.internal.http.HttpEngine.sendRequest(HttpEngine.java:232)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.execute(HttpURLConnectionImpl.java:465)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponse(HttpURLConnectionImpl.java:411)
    at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getInputStream(HttpURLConnectionImpl.java:248)
    at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getInputStream(DelegatingHttpsURLConnection.java:211)
    at com.android.okhttp.internal.huc.HttpsURLConnectionImpl.getInputStream(HttpsURLConnectionImpl.java:30)
    at com.vlu.buggyreader.MainActivity.downloadTitles(MainActivity.kt:48)
```

Note: error 2 says "main" because the uncaught exception is thrown again in the coroutine on main. Full logs: BuggyReader/docs/logcat.